Fikile Mkhize 0210210785080

21 August 2026

EXPLAIN HOW DATA IS STORED ON COMPUTERS

US 14944

FORMATIVE ASSESSMENT

QUESTION 1

1. Bits and bytes is a combination of a bit or binary digit which have value of 0 or 1 and bytes are the standard unit for all computing devices
2. Numeric Data are numbers ,examples are integers which are whole number's, real numbers, currency refers to real numbers formatted in a specific way shown with a symbol and two decimal places, and percentage which is the fractional real numbers formatted in a specific way out of 100.
3. Alphanumeric or text data refers to data made up of letters and numbers
4. Boolean or logical data which has only two values, true or false
5. Floating point data is a variable type that is used to store floating point number values

QQUESTION2

Coding system is method to give short code to represent data for easy identification.

1. American standard code information interchange is used I the transmission of data through data communications and is used to represent data internally in Microcomputers
2. Extended binary coded decimal interchange provides a unique code for each decimal value 0 through 9.
3. Unicode is a worldwide standard where each character uses a unique number
4. Binary coded decimal is that system or code which has the binary numbers or digits to represent a decimal number

Question 3

Data manipulation involves performing calculations on numbers and manipulating strings. Programs manipulate and use variables in many ways .

1. Numeric or arithmetic operators ore operational signs like addition ,subtraction, multiplication, division, integer division and arithmetic operators are Mod modulus arithmetic and strings concatenate
2. Comparison operators are equality, inequality, less than or greater than
3. Logical operators are AND or OR operators

Fikile Mkhize

Question 4

1. Files: a computer file is used for storing information for example a file can contain the payroll information for each employee, with each employee represented by a line
2. Lists : a list contains elements of one particular data type for example a list could contain strings ,names of all the players on a soccer team
3. Array is a structure where the elements are identified by one or more indices and has multiple dimensions
4. Stacks : a stack is a data structure where elements are only inserted or deleted at the top
5. Databases: a database is a structured collection of data

Question 5

1. Master File

- Permanent file, holds main up-to-date information about business entities. Updated periodically. Reference file.

- Example: Customer Master File holds all customer details (name, cell, balance) - at Greatermans Salon: all registered customers. Product Master File - all hair products with current stock level. Employee Master File.

- Nature: Static, permanent.

2. Transaction File

- Temporary file, holds daily transactions/changes that will be used to update master file.

- Example: Daily sales transactions for 08/08/2026 - each sale record (time, item, amount). Daily complaint log file - today's 3 new complaints before merging to master complaint file.

- Nature: Dynamic, moves to master at end of day.

3. Serial File

- Records stored in order they arrive, no specific order. Chronological.

- Example: Unsorted complaint forms as they arrive dropped in box. Log of calls received in order of time - 08:00 call, 08:05 call, 08:10 call.

- Access: Must read from start to find item.

4. Sequential File

- Records sorted by key field (e.g., by ID or surname alphabetical). Must be accessed sequentially from start.

Fikile Mkhize

- Example: Employee file sorted by Employee Number 001,002,003. Customer file sorted by surname A-Z: Dlamini, Molefe, Nkosi. Invoice file sorted by date.

- Access: Must read one by one, but in sorted order.

5. Indexed File / Indexed Sequential

- Has index (like book index) to jump directly to record without reading all. Has primary key + index table.

- Example: Customer file with index on Customer ID - to find GR-002, system uses index to go directly. Like dictionary index. Used in salon CRM - search by cell number using index.

- Access: Direct via index + sequential.

6. Direct / Random / Relative File

- Records stored with address calculated by formula (hashing). Can access any record directly without reading others.

- Example: Employee file where record 10 stored at position 10 on disk. RAM memory access - go directly to address 0x1234. Booking system - go directly to appointment slot 10:00 without checking 09:00.

- Access: Fastest, direct.

Question 6

Differences Table

Feature	Hierarchical Network	Relational
Structure	Tree, one parent	Graph, many parents
Relationship	One to many	Many to many
Example	File system	Student-Subject
Flexibility	Low	Medium